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6 Agricultural and Forest Entomology

7 Royal Entomological Society

8 **RE:** Submission of a Research Article to Agricultural and Forest Entomology

9 Dear Editors,

10 I submit the attached manuscript, "Testing the wind disruption and disturbance refugia
11 hypothesis for mountain pine beetle outbreaks (*Dendroctonus ponderosae*). Terrain, stand
12 density and flight-period wind as controls on attack in the Selkirk Mountains of British Columbia",
13 to Agricultural and Forest Entomology as a Research Article.

14 Krawchuk et al. (2020) proposed that stands escape mountain pine beetle outbreaks
15 where shade, low host density and few large hosts make them refugia, and that wind breaks up
16 the aggregation pheromone in thin stands. This paper tests those mechanisms across 5,573 ha
17 over eight outbreak years, with terrain-resolved sixteen-day wind and previous-year attack in
18 every model.

19 It grew from our previous regeneration study on the same ground (Murphy et al. 2026)ⁱ,
20 which found that ruggedness shaped regeneration but not which terrain property the beetle
21 answered to. It reuses those ground plots to map attack and separates ruggedness into the
22 components the beetle's biology names.

23 Flight-period wind weakens the advantage a dense canopy gives the beetle, the plume
24 mechanism the hypothesis anticipated, while sheltered open ground receives more wind-borne
25 beetles whatever the stand, which it did not. For managers, refugia cannot be mapped from terrain
26 alone, and thinning should lower attack on windward ground in windy flight periods but not on lee
27 slopes.

28 The manuscript is within the 8,000-word limit, with the MicroMet equations, three
29 descriptive tables and a wind-window sensitivity analysis in Supporting Information. The work is
30 original, unfunded and not under consideration elsewhere. I am the sole author, declare no conflict
31 of interest or dispute over the data, have acknowledged every contribution, and have made the
32 derived data and full code public on GitHub.

34 Yours sincerely,

35 Dr Seamus Murphy

ⁱ Murphy, S., Leslie, A., Wilson, J., & Banks, L. K. (2026). Spatial patterns of conifer regeneration following mountain pine beetle outbreak (*Dendroctonus ponderosae*) and mixed-severity wildfire: A point process model of disturbance interactions in the southern Selkirk Mountains, British Columbia. *Forest Ecology and Management*, 618, 123985. <https://doi.org/10.1016/j.foreco.2026.123985>
